

PROJECT PLANNING & MILESTONE SCHEDULE

Development Timeline, Risk Assessment, and Resource Allocation Plan

1. Milestones & Phases

- **Phase 1 (Database & Setup)**: Set up SQLite schema and populate mock concepts. Duration: 3 days.
- **Phase 2 (AI Engine & Audio Logic)**: Develop Whisper transcription, Librosa audio feature extraction, and Sentence-BERT similarity modules. Duration: 5 days.
- **Phase 3 (Dashboard & Presentation)**: Construct Streamlit UI, load custom CSS stylesheets, and design ReportLab PDF templates. Duration: 4 days.
- **Phase 4 (Validation & Optimization)**: Run performance benchmarks, implement model caching, and test edge cases. Duration: 3 days.

2. Risk Analysis & Mitigation

- **Risk 1: Missing ffmpeg Dependency on Target Windows Machine**

Impact: High. Brakes Whisper's default file loader.

Mitigation: Bypassed ffmpeg by using Librosa to read audio files directly and passing pre-processed numpy arrays to Whisper's transcribe function.

- **Risk 2: Model Loading Latency**

Impact: Medium. Reloading heavy models on every click breaks UI responsiveness.

Mitigation: Wrapped model loading in Streamlit's `@st.cache_resource` caching wrapper.